# R workshop part 5 - Paths, projects, and data import

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# Paths

Paths are directions that tell R where to look for data or place outputs on our computer.

- Example Windows path: `C:\Users\gacnik\Desktop`
- Example Mac path: `/Users/gacnik/Desktop`

R only accepts paths with forward slash character (`/`), meaning that:

- Windows paths cannot be used as is, replacement with `/` characters
- Mac paths can be used as is

# Paths

Use `getwd()` to **get** your *current* R working directory. This is where R currently looks for data and saves outputs to.

```
getwd()
```

```
[1] "C:/Users/gacnik/OneDrive - ijs.si/Jan 2023+/R course/Data_analysis_in_R"
```

Use `setwd()` to **set** your *new* R working directory.

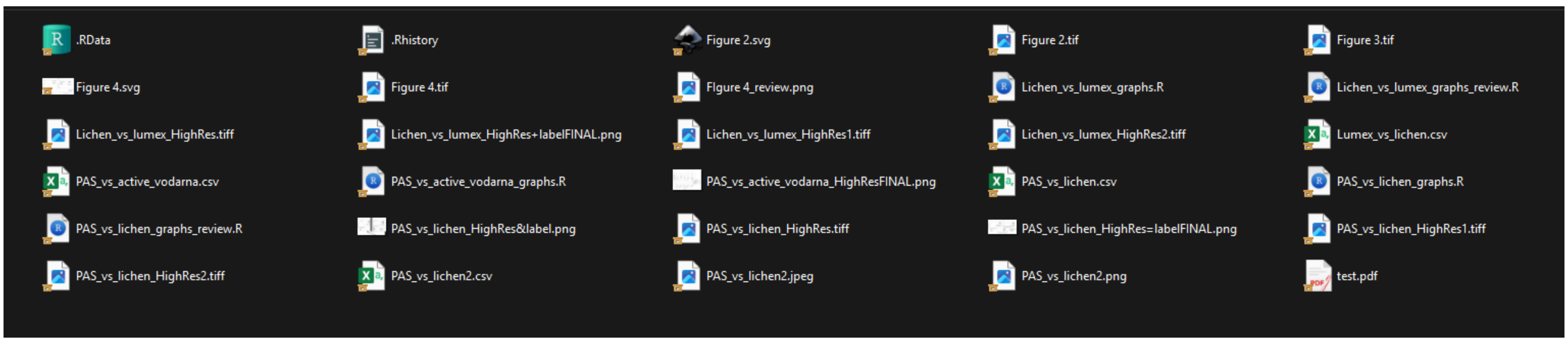
```
setwd("C:/ENTER/YOUR/PATH/HERE")
```

There is an easier and better way to manage paths through **RStudio projects**

# Organizing your analysis project

As your data analysis skill improve, tasks become larger, and supporting files start growing.

- We started our work in an R script, then we generated some plots, tables… if we started saving these outputs and adding new data, things could get disorganized fast:




Personal archive, my first messy R project

# Organizing your analysis project

Project organization for larger tasks/projects:

- everything in **one project folder**, with standard **sub folders**
  - subfolders `data`, `scripts`, `outputs`, …
- use of **RStudio project file** (`.Rproj` file)
  - relative instead of absolute paths
  - preserves specific rstudio environment and setup just for the project

(Showcase the use of RStudio project)

# Relative paths make projects portable

We said RStudio project uses relative paths, not absolute, what does it mean?

- **Absolute path:** starts from a fixed location on one computer
  `C:/Users/gacnik/Desktop/Example_project/data/penguins.csv`
- **Relative path:** starts from the project folder
  `data/penguins.csv`

**Rule:** Always use relative paths in project scripts. Avoid hard-coded absolute paths—they usually break when a project is moved, shared, or opened on another computer.

# Data import

Working with data already provided by R was neat, but you will need to apply the learned concept on your data.

Here, we will focus on plain-text rectangular files such as CSV, TSV, and Excel files

```
Code,Station name,Units,Observed conc,Modelled conc
DE0002R,Waldhof,ng/m3,1.58,1.23
DE0003R,Schauinsland,ng/m3,1.24,1.21
DE0009R,Zingst,ng/m3,1.45,1.29
EE0009R,Lahemaa,ng/m3,1.10,1.25
FI0027R,Virolahti Harju,ng/m3,1.16,1.22
FI0050R,Hyytiala,ng/m3,1.24,1.18
GB0048R,Auchencorth Moss,ng/m3,1.29,1.22
GB1055R,Chilbolton Observatory,ng/m3,1.42,1.28
NO0042G,Zeppelin mountain ,ng/m3,1.53,1.13
NO0073U,Sofienbergparken,ng/m3,1.50,1.18
PL0005R,Diabla Gora,ng/m3,1.37,1.19
SE0014R,Rao,ng/m3,1.26,1.26
```

Example of a simple CSV file

# Data import

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| Code | Station name | Units | Observed conc | Modelled conc |
|---|---|---|---|---|
| **DE0002R** | Waldhof | ng/m3 | 1.58 | 1.23 |
| **DE0003R** | Schauinsland | ng/m3 | 1.24 | 1.21 |
| **DE0009R** | Zingst | ng/m3 | 1.45 | 1.29 |
| **EE0009R** | Lahemaa | ng/m3 | 1.1 | 1.25 |
| **FI0027R** | Virolahti Harju | ng/m3 | 1.16 | 1.22 |
| **FI0050R** | Hyytiala | ng/m3 | 1.24 | 1.18 |
| **GB0048R** | Auchencorth Moss | ng/m3 | 1.29 | 1.22 |
| **GB1055R** | Chilbolton Observatory | ng/m3 | 1.42 | 1.28 |
| **NO0042G** | Zeppelin mountain | ng/m3 | 1.53 | 1.13 |
| **NO0073U** | Sofienbergparken | ng/m3 | 1.5 | 1.18 |
| **PL0005R** | Diabla Gora | ng/m3 | 1.37 | 1.19 |
| **SE0014R** | Rao | ng/m3 | 1.26 | 1.26 |

Same example, viewed as a table

# Import data with a relative path

Download and extract the prepared Example RStudio project.

Open `Example_project.Rproj`. Then import the file with `read_csv()` using a relative path:

```
library(tidyverse)

mercury_air <- read_csv("data/mercury_air.csv")
```